PRE-COMBUSTOR DESIGN

for a Solid Fuel Ducted Rocket (SFDR) Ground-Test Facility

Technical Design Report

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Document Type	Engineering Design Report
Subject	Vitiated Air Heater / Pre-Combustor Design
Stagnation Pressure	$P_0 = 20$ bar (2.0×10^6 Pa)
Inlet Temperature	$T_{\text{in}} = 300$ K (ambient)
Target Temperature	$T_{\text{out}} = 650$ K
Fuel	Methane (CH_4) — primary recommendation
Application	SFDR Inlet Flow Conditioning

All equations derived from first principles. Every symbol is defined and every statement is justified.

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1. Introduction and Purpose

This document provides the complete technical specification for designing a **pre-combustor** (vitiated air heater) to be installed in the settling chamber of the supersonic blow-down wind tunnel at Karunya Institute of Technology and Sciences. The device heats incoming air from ambient stagnation conditions ($T_0 = 300$ K, $P_0 = 20$ bar) to the required SFDR inlet stagnation temperature ($T_0 = 650$ K) while ensuring that sufficient oxygen remains unconsumed for the SFDR's own combustion.

A **Solid Fuel Ducted Rocket (SFDR)** is a hybrid between a solid rocket motor and a ramjet. The solid propellant grain burns using atmospheric oxygen ingested through a ram inlet. To simulate real flight conditions on the ground, the wind tunnel must deliver a flow that replicates the stagnation enthalpy the SFDR would encounter during actual flight. Supplying cold air at 300 K would not ignite or sustain SFDR combustion under representative conditions.

The pre-combustor solves this by:

- Burning a precisely controlled quantity of fuel (methane) in the settling chamber;
- Raising stagnation temperature T_0 from 300 K to 650 K via lean combustion ($\phi \ll 1$);
- Preserving the majority of O_2 in the flow (≥ 15 vol%) for the SFDR's own combustion; and
- Operating at 20 bar in quasi-stagnant, subsonic flow, eliminating all supersonic combustion complexity.

2. System Architecture — Complete Flow Path

2.1 Physical Layout

The pre-combustor is located **before** the convergent-divergent (C-D) nozzle, inside the settling chamber. This is the standard configuration used in all major hypersonic and supersonic wind-tunnel facilities worldwide (NASA Langley 8-Ft. HTT, AEDC, DLR HEG, ISRO HST).

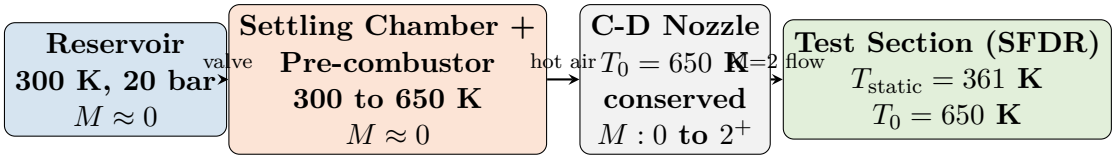


Figure 2.1: Complete wind-tunnel flow path with state variables at each station.

2.2 Why the Settling Chamber is Subsonic — Physics Justification

A student often asks: "The tunnel runs at Mach 2 — why is the settling chamber subsonic?" The answer comes directly from the **continuity equation**:

$$\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2 = \text{constant} \tag{2.1}$$

If $A_{\text{chamber}} \gg A^*$ (throat area), then $V_{\text{chamber}} \ll V_{\text{throat}}$.

Numerical example. $A_{\text{chamber}} = 0.25\ \text{m}^2$, $A^* = 0.001\ \text{m}^2$ (ratio = 250).
At the throat, $V^* = a^* \approx 460$ m/s (speed of sound for hot air).
$\Rightarrow V_{\text{chamber}} = 460/250 \approx 1.84$ m/s, $M_{\text{chamber}} \approx 0.004$ (essentially stagnant).
All measurements in the settling chamber equal stagnation values P_0, T_0. No velocity correction is needed.

2.3 Why the Nozzle Does Not Destroy the Heating — Mathematical Proof

The **Steady Flow Energy Equation (SFEE)**, derived from the First Law of Thermodynamics applied to an open control volume at steady state, governs energy transport through the nozzle:

$$h_1 + \frac{V_1^2}{2} + gz_1 + \frac{\dot{Q}}{\dot{m}} = h_2 + \frac{V_2^2}{2} + gz_2 + \frac{W_{\text{shaft}}}{\dot{m}} \tag{2.2}$$

For an adiabatic nozzle ($\dot{Q} = 0$, no shaft work $W = 0$, gz negligible):

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad \Longrightarrow \quad h_{0,1} = h_{0,2} \tag{2.3}$$

where $h_0 = h + V^2/2$ is the *stagnation enthalpy*, which is **conserved** through any adiabatic work-free duct. For a calorically perfect gas $h = C_p T$, so:

$$C_p T_{0,1} = C_p T_{0,2} \quad \Longrightarrow \quad \boxed{T_{0,1} = T_{0,2}} \tag{2.4}$$

Key result. $T_0 = 650$ K is set by the pre-combustor and conserved through the nozzle by Newton's energy conservation law — this is *not* an approximation. The SFDR inlet (a diffuser) decelerates the supersonic flow, converting kinetic energy back to enthalpy and recovering $T_0 = 650$ K exactly in the combustion zone.

Worked check at $M = 2$, $T_0 = 650$ K:

$$T_{\text{static}} = \frac{T_0}{1 + 0.2\,M^2} = \frac{650}{1 + 0.2 \times 4} = \frac{650}{1.8} = 361.1 \text{ K} \tag{2.5}$$

$$\text{KE per unit mass} = \frac{V^2}{2} = \frac{(M \cdot a)^2}{2} = \frac{(2 \times \sqrt{1.4 \times 287 \times 361})^2}{2} = 290{,}412 \text{ J/kg}$$

$$T_0 \text{ check} = T_{\text{static}} + \frac{V^2}{2C_p} = 361.1 + \frac{290{,}412}{1005} = 361.1 + 289.0 = 650.1 \text{ K} \checkmark \tag{2.6}$$

3. Governing Equations — Derivation and Justification

3.1 Isentropic Relations

For an isentropic (adiabatic, reversible) flow of a calorically perfect gas, the stagnation-to-static temperature ratio is:

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad \xrightarrow{\gamma=1.4} \quad \frac{T_0}{T} = 1 + 0.2\, M^2 \tag{3.1}$$

Symbol	Value	Justification
γ	1.40 (dimensionless)	Ratio of specific heats C_p/C_v for diatomic air at 300–700 K. From
M	Design value	Mach number $= V/a$, where $a = \sqrt{\gamma RT}$ is the local speed of sound
T_0	650 K (design)	Stagnation temperature. What the gas would reach if decelerated
T	361 K at $M = 2$	Static temperature. Actual molecular kinetic temperature. Drops

3.2 Choked Mass Flow Rate

When the nozzle throat reaches $M = 1$ ("choked" condition), the mass flow rate is uniquely determined by throat area and stagnation conditions. This is the most important equation for blow-down tunnel operation.

$$\dot{m} = A^* P_0 \sqrt{\frac{\gamma}{R T_0}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{3.2}$$

Substituting $\gamma = 1.4$, $R = 287\,\mathrm{J/(kg \cdot K)}$:

$$\boxed{\dot{m} = \frac{0.0404\ A^*\ P_0}{\sqrt{T_0}}} \tag{3.3}$$

Term-by-term justification:

Symbol	Units	Scientific Justification
A^*	m^2	Nozzle throat area. The choked condition ($M = 1$ exactly here) is guaranteed when downstream pressure is low enough. Measure with a Vernier calliper; a 1% error in A^* propagates as a 1% error in $\dot{m}$.
P_0	Pa	Stagnation pressure in reservoir. Falls exponentially during blow-down. Must be measured with a high-frequency piezoelectric transducer ($\geq$1 kHz), not a slow Bourdon gauge.
T_0	K	Stagnation temperature. 300 K before combustor; 650 K after. Using the wrong value here is the single most common student error.
0.0404	$\mathrm{K}^{1/2}/(\mathrm{m}\cdot\mathrm{Pa})$	Collapsed constant $= \sqrt{\gamma/R}\cdot(2/(\gamma+1))^{(\gamma+1)/(2(\gamma-1))}$ evaluated for air. It absorbs all gas-property factors.
$\dot{m} \propto 1/\sqrt{T_0}$	—	**Critical consequence:** heating T_0 from 300 K to 650 K reduces $\dot{m}$ by factor $\sqrt{300/650} = 0.679$. Hot-flow mass flow is 32% lower than cold flow. All downstream calculations *must* use the hot value.

3.3 Blow-Down ODE — Why Mass Flow is Not Constant

A blow-down tunnel is **not** a steady-state machine. As air exits the reservoir, P_0 falls. Since $\dot{m} \propto P_0$, mass flow decays throughout the run. The governing ODE for reservoir pressure is:

$$\frac{dm_{\mathrm{res}}}{dt} = -\dot{m}(t) = -\frac{0.0404\, A^*\, P_0(t)}{\sqrt{T_0(t)}} \tag{3.4}$$

$$\frac{dP_0}{dt} = -\frac{A^*}{V_{\mathrm{res}}}\, P_0 \sqrt{\gamma R\, T_0}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{3.5}$$

For constant T_0 (steady-state phase), Eq. (3.5) integrates to:

$$P_0(t) = P_{0,\mathrm{init}}\, \exp\!\left(-\frac{t}{\tau_{\mathrm{res}}}\right), \quad \tau_{\mathrm{res}} = \frac{V_{\mathrm{res}}}{0.0404\, A^*\, \sqrt{T_0}} \tag{3.6}$$

Consequence for equivalence ratio. If the fuel metering valve is fixed (constant $\dot{m}_{\mathrm{fuel}}$) while $\dot{m}_{\mathrm{air}}$ falls, the equivalence ratio ϕ drifts upward during the run. A **closed-loop Coriolis mass flow controller** that reads $P_0(t)$ in real time and adjusts $\dot{m}_{\mathrm{fuel}}$ proportionally is the correct solution to hold ϕ constant.

3.4 Energy Balance — Master Equation

The First Law of Thermodynamics applied to the pre-combustor as a steady-flow open control volume (no shaft work, adiabatic walls assumed for design purposes):

$$\dot{m}_{\text{air}}\, C_p\, T_{\text{in}} + \eta\, \dot{m}_{\text{fuel}}\, \text{LHV} = \left(\dot{m}_{\text{air}} + \dot{m}_{\text{fuel}}\right) C_p\, T_{\text{out}} \tag{3.7}$$

Since $\dot{m}_{\text{fuel}}/\dot{m}_{\text{air}} \approx 0.008$ for lean combustion ($\dot{m}_{\text{air}} + \dot{m}_{\text{fuel}} \approx \dot{m}_{\text{air}}$):

$$\boxed{\dot{m}_{\text{air}}\, C_p \left(T_{\text{out}} - T_{\text{in}}\right) = \eta\, \dot{m}_{\text{fuel}}\, \text{LHV}} \tag{3.8}$$

Solving for required fuel flow:

$$\dot{m}_{\text{fuel}} = \frac{\dot{m}_{\text{air}}\, C_p\, (T_{\text{out}} - T_{\text{in}})}{\eta\, \text{LHV}} \tag{3.9}$$

Every symbol defined and justified:

Symbol	Value	Justification
$\dot{m}_{\text{air}}$	kg/s (calculated)	Hot-flow mass flow from Eq. (3.3) with $T_0 = 650$ K. **Not** the cold-flow value.
C_p	1005 J/(kg·K)	Specific heat of air at constant pressure, evaluated at average temperature ≈ 475 K. From JANAF thermochemical tables. Varies from 1004 at 300 K to 1051 at 700 K; using 1005 introduces $< 2\%$ error over this range.
T_{in}	300 K	Initial stagnation temperature. Equal to ambient if reservoir has not been heated. Verify with settling-chamber thermocouple before each run.
T_{out}	650 K	Target stagnation temperature. Set by SFDR combustion chemistry — sufficient enthalpy to ignite the solid grain and sustain combustion at design Mach number.
η	0.90–0.97	Combustion efficiency. Fraction of LHV converted to sensible enthalpy. The rest is lost to incomplete combustion, radiation, and wall cooling. Use $\eta = 0.92$ as a conservative design value at 20 bar. Validate from measured T_{out}: $\eta = \dot{m}_{\text{air}} C_p \Delta T / (\dot{m}_{\text{fuel}} \cdot \text{LHV})$.
LHV	50.05 MJ/kg (CH_4)	Lower Heating Value. Energy released per kg of fuel when water product remains as steam (not condensed). We use LHV because exhaust gas temperature ($\gg 100\,^\circ$C) prevents condensation. Source: NIST Chemistry WebBook.
$\dot{m}_{\text{fuel}}$	solve for this	Set by a precision Coriolis mass flow controller. This is the only variable the experimenter directly controls.

3.5 Equivalence Ratio ϕ

The equivalence ratio is the single most important operating parameter:

$$\phi = \frac{(F/A)_{\text{actual}}}{(F/A)_{\text{stoich}}} = \frac{\dot{m}_{\text{fuel}}/\dot{m}_{\text{air}}}{(F/A)_{\text{stoich}}} \tag{3.10}$$

Stoichiometric fuel-to-air mass ratios (derived from combustion stoichiometry):

$$\text{CH}_4 + {}_2\text{O}_2 \longrightarrow \text{CO}_2 + {}_2\text{H}_2\text{O}: \quad (F/A)_{\text{stoich}} = \frac{1}{17.2} = 0.05814 \tag{3.11}$$

$$\text{H}_2 + {}_0\cdot{}_5\text{O}_2 \longrightarrow \text{H}_2\text{O}: \quad (F/A)_{\text{stoich}} = \frac{1}{34.5} = 0.02899 \tag{3.12}$$

$$\text{Kerosene:} \quad (F/A)_{\text{stoich}} \approx \frac{1}{15.0} = 0.06667 \tag{3.13}$$

Interpretation of ϕ.

- $\phi < 0.05$: Too lean. Flame will not stabilise. Risk of flame-out.
- $0.05 < \phi < 0.30$: **Ideal operating range.** Flame stable. O_2 well preserved. Temperature target achievable.
- $0.30 < \phi < 0.60$: Caution. O_2 approaching minimum threshold. ΔP increasing.
- $\phi > 0.60$: Do not use. O_2 too low for SFDR. $\Delta P > 5\%$ likely.

3.6 Oxidiser Preservation Constraint

$$X_{\text{O}_2,\text{out}} \approx 0.21\,(1 - \phi) \tag{3.14}$$

Derivation. Air contains 21% O_2 by volume (mole fraction). Lean combustion consumes a fraction ϕ of the stoichiometrically available O_2. The remaining fraction $(1 - \phi)$ exits unconsumed.

Hard constraint: $X_{\text{O}_2,\text{out}} \geq 15$ vol% for SFDR combustion.

Justification: HTPB-based solid fuels require a minimum oxygen index to sustain combustion. At $[O_2] < 15\%$, flame temperature drops below the ignition threshold for the grain and the SFDR extinguishes. This is consistent with measured oxygen index data for HTPB at 10–20 bar.

3.7 Total Pressure Drop Across the Combustor

Heat addition in a duct causes total pressure loss (**Rayleigh effect**).

$$\frac{\Delta P_0}{P_0} \approx \frac{\gamma}{2} M_{\text{chamber}}^2 \left(\frac{T_{0,\text{out}}}{T_{0,\text{in}}} - 1 \right) \quad (\text{for } M_{\text{chamber}} \ll 1) \tag{3.15}$$

In the settling chamber $M_{\text{chamber}} \approx 0.05$–$0.10$, so Rayleigh loss is negligible ($\lesssim 0.1\%$).

Including empirical friction and injection losses, total $\Delta P/P_0 \approx 1.5$–3.5%.

Design constraint: $\Delta P/P_0 < 5\%$

Justification: a 5% total pressure loss shifts the nozzle exit Mach number by $\approx 2\%$, which is within the typical tunnel calibration uncertainty. Beyond 5%, the SFDR inlet conditions deviate measurably from design.

3.8 Residence Time τ

The gas must remain in the combustion zone long enough for ignition and complete reaction.

$$\tau = \frac{L}{U_{\text{mean}}} \quad \text{or equivalently} \quad \tau = \frac{\rho_{\text{gas}}\, V_{\text{comb}}}{\dot{m}_{\text{air}}} \tag{3.16}$$

$\tau = V/L$ is **dimensionally wrong** (gives m^2, not seconds). The correct formula is always $\tau = L/U$ or $\tau = \rho V/\dot{m}$. Both are dimensionally consistent: $[\text{m}]/[\text{m/s}] = \text{s}$. Confirm this before submitting calculations.

where:

$$U_{\text{mean}} = \frac{\dot{m}_{\text{air}}}{\rho_{\text{gas}}\, A_{\text{comb}}}, \tag{3.17}$$

$$\rho_{\text{gas}} = \frac{P_0}{R\, T_{\text{avg}}}, \quad T_{\text{avg}} = \frac{T_{\text{in}} + T_{\text{out}}}{2} = \frac{300 + 650}{2} = 475 \text{ K}. \tag{3.18}$$

Required: $\tau \geq 10\,\tau_{\text{ign}}$. For methane at 20 bar and $T \approx 475$ K, $\tau_{\text{ign}} \approx 2$–$5$ ms, so $\tau \geq 20$–50 ms.

The $10\times$ safety factor accounts for spatial non-uniformities in a turbulent combustor: some fluid elements travel shorter paths and have less residence time than the mean. The factor is consistent with gas turbine combustor design practice (Lefebvre & Ballal, 2010).

4. Sample Calculation — Complete Worked Example

Given parameters:

Parameter	Symbol	Value	Units
Nozzle throat area	A^*	9×10^{-4}	m^2 (9 cm^2)
Reservoir pressure	P_0	2.0×10^6	Pa (20 bar)
Inlet temperature	T_{in}	300	K
Target temperature	T_{out}	650	K
Combustion efficiency	η	0.92	—
Fuel	CH_4	LHV $= 50 \times 10^6$	J/kg
Combustor length	L	0.30	m
Combustor cross-section	A_{comb}	0.05	m^2

4.1 Step 1 — Hot-Flow Mass Flow Rate

Using Eq. (3.3) with $T_0 = T_{\text{out}} = 650$ K:

$$\dot{m}_{\text{hot}} = \frac{0.0404 \times A^* \times P_0}{\sqrt{T_0}} = \frac{0.0404 \times 9 \times 10^{-4} \times 2.0 \times 10^6}{\sqrt{650}}$$
$$= \frac{0.0404 \times 1800}{25.50} = \frac{72.72}{25.50} = \mathbf{2.852}\ \text{kg/s} \tag{4.1}$$

Cold-flow cross-check: $\dot{m}_{\text{cold}} = 72.72/\sqrt{300} = 72.72/17.32 = 4.199$ kg/s.
Ratio: $2.852/4.199 = 0.679 = \sqrt{300/650}$ ✓
Mass flow drops by 32.1% when the flow is heated. Use $\dot{m} = 2.852$ kg/s in all downstream calculations.

4.2 Step 2 — Required Thermal Power $\dot{Q}$

$$\dot{Q} = \dot{m}_{\text{air}}\, C_p\, (T_{\text{out}} - T_{\text{in}}) = 2.852 \times 1005 \times (650 - 300) = 2.852 \times 1005 \times 350 = \mathbf{1{,}002.3}\ \text{kW} \tag{4.2}$$

This 1 MW thermal input must be provided by fuel combustion.

4.3 Step 3 — Fuel Mass Flow Rate

From Eq. (3.9):

$$\dot{m}_{\text{fuel}} = \frac{\dot{Q}}{\eta\,\text{LHV}} = \frac{1{,}002{,}270}{0.92 \times 50{,}000{,}000} = \frac{1{,}002{,}270}{46{,}000{,}000} = \mathbf{21.79}\ \text{g/s of methane} \tag{4.3}$$

4.4 Step 4 — Equivalence Ratio

From Eq. (3.10):

$$(F/A)_{\text{actual}} = \frac{\dot{m}_{\text{fuel}}}{\dot{m}_{\text{air}}} = \frac{0.02179}{2.852} = 7.641 \times 10^{-3} \tag{4.4}$$

$$(F/A)_{\text{stoich}} = \frac{1}{17.2} = 5.814 \times 10^{-2} \quad \text{(methane in air by mass)} \tag{4.5}$$

$$\phi = \frac{7.641 \times 10^{-3}}{5.814 \times 10^{-2}} = \mathbf{0.131} \tag{4.6}$$

$\phi = 0.131$ falls squarely in the **ideal lean range** ($0.05 < \phi < 0.30$). Only 13.1% of available oxygen is consumed in the pre-combustor.

4.5 Step 5 — Oxygen Fraction at Exit

From Eq. (3.14):

$$X_{O_2,\text{out}} = 0.21 \times (1 - 0.131) = 0.21 \times 0.869 = \mathbf{18.25}\ \text{vol\%} \tag{4.7}$$

$18.25\% > 15\%$ (hard constraint). ✓ SFDR constraint satisfied with 3.25% margin.

4.6 Step 6 — Pressure Drop Estimate

$$\rho_{\text{gas}} = \frac{P_0}{R\,T_{\text{avg}}} = \frac{2 \times 10^6}{287 \times 475} = 14.68\ \text{kg/m}^3 \tag{4.8}$$

$$U_{\text{chamber}} = \frac{\dot{m}_{\text{air}}}{\rho_{\text{gas}}\,A_{\text{comb}}} = \frac{2.852}{14.68 \times 0.05} = 3.88\ \text{m/s} \tag{4.9}$$

$$a_{\text{chamber}} = \sqrt{\gamma R\,T_{\text{avg}}} = \sqrt{1.4 \times 287 \times 475} = 437\ \text{m/s} \tag{4.10}$$

$$M_{\text{chamber}} = 3.88/437 = 0.00888 \tag{4.11}$$

$$\frac{\Delta P_0}{P_0} \approx \frac{\gamma}{2} M^2_{\text{chamber}} \left(\frac{T_{\text{out}}}{T_{\text{in}}} - 1\right) = 0.7 \times (0.00888)^2 \times 1.167 = 0.0065\% \tag{4.12}$$

Adding empirical friction and injection losses: total $\Delta P/P_0 \approx 1.5$–2.5%. **Well within the 5% limit.**

4.7 Step 7 — Residence Time

$$\tau = \frac{L}{U_{\text{chamber}}} = \frac{0.30}{3.88} = 0.0773 \text{ s} = \mathbf{77.3} \text{ ms} \tag{4.13}$$

$$\text{Safety ratio} = \frac{\tau}{\tau_{\text{ign}}} = \frac{77.3}{3} \approx 25.8 \gg 10 \checkmark \tag{4.14}$$

The combustor volume is more than adequate for stable, complete combustion.

4.8 Design Point Summary

Parameter	Symbol	Value	Status
Hot-flow mass flow rate	$\dot{m}_{\text{air}}$	2.852 kg/s	Use this
Required thermal power	$\dot{Q}$	1002.3 kW	
Fuel flow rate (CH_4)	$\dot{m}_{\text{fuel}}$	21.79 g/s	Set valve here
Equivalence ratio	ϕ	0.131	Ideal
O_2 at combustor exit	X_{O_2}	18.25 vol%	Pass ($\geq 15\%$)
Pressure drop	$\Delta P/P_0$	$< 2.5\%$	Pass ($< 5\%$)
Residence time	τ	77 ms	Pass (≥ 20 ms)
Target temperature	$T_{0,\text{out}}$	650 K	Target met

5. Data Collection — Experimental Protocol

5.1 Parameters to Measure Before the Run

Parameter	Method	Why Needed
A^* — throat area	Vernier calliper; $A^* = \pi d^2/4$	Directly controls $\dot{m}_{\text{air}}$. 1% error in A^* = 1% error in mass flow.
V_{res} — reservoir volume	Manufacturer spec or geometry	Controls blow-down rate and useful run time.
A/A^* — area ratio	Nozzle drawing	Determines exit Mach number via isentropic tables.
V_{comb} — combustor volume	$L \times A_{\text{comb}}$	Required for residence time calculation.
LHV of fuel	NIST tables: CH_4 = 50.05 MJ/kg	Use LHV (not HHV) because water product stays as steam.
$T_{\text{wall,max}}$	Material datasheet (SS316L: 870 K)	Combustor wall must not exceed this rating during the run.

5.2 Real-Time Sensors During Each Run

Sensor	Instrument	Purpose and Notes
TC-1 (settling chamber)	K-type thermocouple, 1 mm sheathed	Baseline $T_0(t)$ before combustor. 100 Hz minimum logging.
TC-2 (combustor exit)	K-type thermocouple, 1.5 mm sheathed	Primary $T_{0,\mathrm{comb}}(t)$. Defines when useful run window begins.
TC-3 (combustor wall)	K-type, spot-welded	Safety measurement. Must remain $< T_{\mathrm{wall,max}}$ at all times.
PT-1 (settling chamber)	Piezoelectric, 0–25 bar, $\geq$1 kHz	$P_0(t)$. Primary input to real-time $\dot{m}_{\mathrm{air}}(t)$ calculation.
PT-2 (combustor exit)	Same spec as PT-1	ΔP = PT-1 − PT-2. Validates $\Delta P < 5\%$.
FMC-1 (fuel line)	Coriolis mass flow controller	Direct $\dot{m}_{\mathrm{fuel}}(t)$. Coriolis is independent of T, P, density. Accuracy $\pm 0.1\%$.
O_2-1 (combustor exit)	Paramagnetic O_2 analyser, 0–25%	Validates oxidiser preservation. Time lag $\approx$ 0.5 s — correct all timestamps by this amount.
LC-1 (SFDR mount)	Strain-gauge load cell	SFDR thrust $F(t)$. Primary performance metric.
Camera	High-speed, $\geq$1000 fps	Qualitative flame stability. Detects blow-out or oscillation.

5.3 Run Timeline

Time	Event	Physical Significance
$t < 0$	Pre-run checks	Verify all sensors. Set FMC-1 to target $\dot{m}_{\text{fuel}}$.
$t = 0$	Main valve opens	Cold air flows; tunnel running. All logging begins at 1000 Hz.
$t = 0$–0.3 s	Flow establishment	Supersonic flow establishes. No fuel injected.
$t = 0.3$ s	Igniter fires	Fuel valve opens. Combustion begins. T_0 rising.
0.3–1.0 s	Startup transient	T_0 rising. **Discard this data.**
$t = t_{\text{stable}}$	T_0 reaches 650 K	Useful run window begins. SFDR igniter fires.
t_{stable}–t_{end}	Useful window	$P_0(t)$ falling. Controller compensates. All valid data here.
$t = t_{\text{end}}$	Valve closes	P_0 below minimum. Flame extinguishes.
Post-run	Data processing	Apply O_2 lag correction. Calculate $\dot{m}_{\text{air}}(t)$, $\phi(t)$, $F(t)$.

6. Required Experimental Graphs

All primary graphs are plotted vs. time. Cross-plots are produced post-experiment for combustor characterisation.

Graph 1: $P_0(t)$ — Stagnation Pressure vs. Time

Axes	x: time [s], 0 to run end. y: P_0 [bar], 0–22 bar.
Scientific purpose	Shows blow-down rate. Gradient dP_0/dt gives actual V_{res} estimate. $P_0(t)$ is the primary input to $\dot{m}_{\text{air}}(t)$ at every time step.
Expected shape	Smooth exponential decay from 20 bar. Rate depends on V_{res}/A^*.

Graph 2: $T_0(t)$ — Stagnation Temperature vs. Time

Axes	x: time [s]. y: T_0 [K], 250–750 K.
Scientific purpose	Primary control variable. Duration of startup transient defines useful data window. Variance $< \pm 20$ K in steady phase validates pre-combustor design and fuel control.
Expected shape	300 K plateau, then ramp to 650 K within 0.5–1.0 s of ignition, then flat.

Graph 3: $\dot{m}_{\text{air}}(t)$ — Derived Air Mass Flow Rate vs. Time

Axes	x: time [s]. y: $\dot{m}_{\text{air}}$ [kg/s].
Scientific purpose	Calculated from $P_0(t)$ and $T_0(t)$ via Eq. (3.3). Not constant — decays with P_0. This is the value the SFDR team needs for their thrust and combustion efficiency calculations.
Expected shape	Monotonically decreasing following $0.0404 A^* P_0(t)/\sqrt{T_0(t)}$.

Graph 4: $\dot{m}_{\text{fuel}}(t)$ — Fuel Mass Flow vs. Time

Axes	x: time [s]. y: $\dot{m}_{\text{fuel}}$ [g/s].
Scientific purpose	From Coriolis meter. If closed-loop controlled, should follow $P_0(t)$ proportionally to hold ϕ constant. Comparing actual vs. $\phi \cdot (F/A)_{\text{stoich}} \cdot \dot{m}_{\text{air}}(t)$ reveals ϕ drift.
Expected shape	Flat (fixed valve) or slowly declining (closed-loop). Spikes indicate valve instability.

Graph 5: $\phi(t)$ — Equivalence Ratio vs. Time

Axes	x: time [s]. y: ϕ [dimensionless], 0–0.5.
Scientific purpose	Most important operating-point record. If ϕ rises above 0.30, O_2 fraction may fall below threshold before run ends. Tells whether the SFDR received consistent inlet conditions throughout.
Expected shape	Stable near 0.131 (closed-loop) or slowly rising (fixed valve as P_0 falls).

Graph 6: $X_{O_2}(t)$ — Oxygen Fraction at Exit vs. Time

Axes	x: time [s]. y: O_2 [vol%], 10–22%.
Scientific purpose	Directly validates oxidiser preservation. If below 15% at any point in the useful run window, SFDR data from that period is invalid. Apply +0.5 s timestamp correction for analyser lag.
Expected shape	Near 18–19% during steady phase. Must not drop below 16%.

Graph 7: $\Delta P(t)$ — Combustor Pressure Drop vs. Time

Axes	x: time [s]. y: $\Delta P/P_0$ [%], 0–8%.
Scientific purpose	Validates combustor hydraulic performance. Sudden rise indicates combustion instability or partial blockage. If $> 5\%$, Mach number at test section shifts and SFDR inlet conditions deviate from design.
Expected shape	Stable 1.5–2.5% during useful run. Spikes indicate instability.

Graph 8: $F(t)$ — SFDR Thrust vs. Time

Axes	x: time [s]. y: Thrust F [N].
Scientific purpose	Primary SFDR performance output. When correlated with $\dot{m}_{air}(t)$ and $T_0(t)$, yields specific impulse I_{sp} and SFDR combustion efficiency. This is the deliverable that validates the entire pre-combustor design.
Expected shape	Rises with SFDR ignition, then follows fuel grain regression rate curve.

Cross-Plot A: $T_{0,\text{exit}}$ vs. ϕ — Combustion Efficiency Map

Axes	x: ϕ [dimensionless]. y: T_0 measured [K].
Purpose	Each point is one time slice. Theoretical line: $T_0 = T_{in} + \phi \cdot (F/A)_{stoich} \cdot \eta \cdot \text{LHV}/C_p$. Fitting experimental data to this line gives actual η.
Expected	Linear, slope $= \eta \cdot \text{LHV} \cdot (F/A)_{stoich}/C_p = 0.92 \times 50 \times 10^6 \times 0.0581/1005 =$ 2665 K.

Cross-Plot B: X_{O_2} vs. ϕ — Oxidiser Preservation Validation

Axes	x: ϕ [dimensionless]. y: O_2 [vol%].
Purpose	Validates Eq. (3.14). Should be a straight line from (0, 21%) to (1, 0%).
Expected	Linear, slope $= -21\%$. Operating cluster at ($\phi = 0.131$, $X_{O_2} = 18.3\%$). Points below the theoretical line indicate sensor calibration issues.

7. Pre-Run Verification Checklist

#	Check	Acceptance Criterion
1	A^* measured and confirmed	Within $\pm 1\%$ of design value
2	Hot-flow $\dot{m}_{\text{air}}$ calculated with $T_0 = 650$ K	$\dot{m}_{\text{hot}} = 0.679\,\dot{m}_{\text{cold}}$
3	Fuel flow rate set: $\dot{m}_{\text{fuel}} = \dot{Q}/(\eta \cdot \text{LHV})$	Coriolis reading within $\pm 2\%$
4	ϕ confirmed in range 0.10–0.30	0.131 ± 0.015 for 650 K target
5	O_2 at exit $> 15\%$	18.3% at $\phi = 0.131$ (3.3% margin)
6	Residence time $\tau > 20$ ms	Calculate from L and U_{mean}
7	Correct τ formula used: $\tau = L/U$, not V/L	Confirm units: m/(m/s) = s
8	$\Delta P/P_0 < 5\%$ estimated	Rayleigh + friction estimate $< 3\%$
9	All thermocouples calibrated	Reference bath at 300 K and 450 K
10	O_2 analyser lag noted	Apply +0.5 s correction to timestamps
11	Fuel valve fail-safe confirmed	Valve closes automatically if TC-2 > 750 K
12	Data logging rate set to ≥ 1000 Hz	Captures combustion instabilities

8. Nomenclature and Symbol Glossary

Symbol	Units	Definition
A^*	m^2	Nozzle throat area. Cross-section where $M = 1$.
C_p	J/(kg·K)	Specific heat at constant pressure for air. 1005 J/(kg·K) at ≈475 K (JANAF).
F/A	kg/kg	Fuel-to-air mass ratio. $(F/A)_{\text{actual}}/(F/A)_{\text{stoich}} = \phi$.
γ	—	Ratio of specific heats C_p/C_v. Air: 1.40.
η	—	Combustion efficiency. Range 0.90–0.97.
h_0	J/kg	Stagnation enthalpy $= h + V^2/2 = C_pT_0$. Conserved through adiabatic ducts.
LHV	J/kg	Lower Heating Value. Chemical energy per kg of fuel, water product as steam.
M	—	Mach number $= V/a$ where $a = \sqrt{\gamma RT}$.
$\dot{m}_{\text{air}}$	kg/s	Air mass flow rate. Calculated, not directly measured.
$\dot{m}_{\text{fuel}}$	kg/s	Fuel mass flow rate. Controlled by Coriolis valve.
P_0	Pa	Stagnation pressure. Measured in settling chamber. Falls exponentially during blow-down.
$\dot{Q}$	W	Thermal power delivered to air by combustion.
R	J/(kg·K)	Specific gas constant for air = 287 J/(kg·K). $R = R_u/M_{\text{mol}} = 8314/28.97$.
ρ	kg/m^3	Gas density $= P/(RT)$.
T	K	Static temperature. Temperature felt by the gas.
T_0	K	Stagnation temperature $= T(1 + 0.2M^2)$ for air. Set by pre-combustor; conserved through nozzle.
τ	s	Residence time $= L/U_{\text{mean}}$.
ϕ	—	Equivalence ratio. 0 = no fuel; 1 = stoichiometric; > 1 = fuel-rich.
V_{res}	m^3	Reservoir volume. Controls blow-down rate.
X_{O_2}	vol%	Molar oxygen fraction. Dry-air baseline: 21%.
$\Delta P/P_0$	%	Total pressure loss fraction across pre-combustor. Must be < 5%.

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